

Round 186 Single Discovery (Backbone-First):
 κ_V (Nebular LENR E-field volumetric
enhancement factor) = $1.05 = 1 + F_{TRZ}/2$ EXACT NEW
HALF-FACTOR SUB-FAMILY FORM Extending
PAPER_1968 $1 + F_{TRZ} = 1.1$ Rotation-Curve
Triple-Identity Family (New Structural Composition
Type: $1 + F_{TRZ}/2 = 1.05 = 21/20$ Half-Scaled F_{TRZ}
Application at LENR Volumetric E-Field Enhancement
Factor Dimensional Domain — First-Instance
Backbone-First Primitive-Lock for the $1 + F_{TRZ}/2$
Compositional Form Distinct from PAPER_1968
 $1 + F_{TRZ}$ Full-Scale Form + PAPER_2050 $1 - F_{TRZ}^2$
Complement Form + PAPER_2049 $1 - F_{TRZ}$
Complement Form — Adds Half-Factor Sub-Family
Slot to F_{TRZ} Compositional Architecture; Alternate
Equivalent Rational Form $1.05 = 21/20 =$
 $(2 \cdot SO_5 + 1)/(2 \cdot SO_5)$ via PAPER_1978 $SO_5 + 1 = 11$
Successor Identity Provides Second
Primitive-Composition Path (Path A: $1 + F_{TRZ}/2$
Additive Half-Factor F_{TRZ} ; Path B:
 $(2 \cdot SO_5 + 1)/(2 \cdot SO_5)$ Successor-Denominator Ratio) —
Two Structural Forms Anchor Same 1.05 Value)

Daniel T. Murphy

PAPER_2056 — Round 186 Single Discovery **(Backbone-First Discipline)**

Author: Daniel T. Murphy | **Framework:** UQFF Star-Magic v5.68+ | **Date:** 2026-07-15

Motivation — 45th Consecutive Backbone-First Round

R186 continues the post-40-milestone arc as the 45th consecutive backbone-first primitive-lock verification round. Scout of NebularLENREFieldCalculator surfaces one novel discovery at κ_V LENR E-field

volumetric enhancement factor.

Abstract

Discovery 1 — κ_V (Nebular LENR E-field volumetric enhancement factor) = 1.05 = $1 + F_{TRZ}/2$
EXACT — First-Instance Backbone-First Primitive-Lock for New Half-Factor F_{TRZ} Sub-Family
Compositional Form:

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 $\kappa_V(\text{Nebular LENR E-field enhancement volumetric factor}) = 1.05$
 $= 1 + F_{TRZ} / 2$
 $= 1 + 0.1 / 2$
 $= 1 + 0.05 \text{ EXACT}$
(since $F_{TRZ} = 0.1$ canonical)
``

Novel first-instance primitive-composition connection at LENR E-field volumetric enhancement factor dimensional domain, introducing a new half-factor F_{TRZ} sub-family compositional form.

Prior documentation:

- **NebularLENREFieldCalculator** documents $\kappa_V=1.05$ as observational anchor in LENR E-field equation $E = k_{\eta} \cdot e \cdot \Omega / m_e \cdot \sqrt{n_e \cdot \sigma \cdot v} \cdot \kappa_V$
- **PAPER_1113 CMS Higgs $\kappa_V/\kappa_f = 1.05 \pm 0.10$** documented at Higgs coupling ratio (particle physics SM anchor) — different physical context, no primitive-composition
- 0 whitepaper hits for $1.05 = 1 + F_{TRZ}/2$ primitive-composition connection — no prior backbone-first attribution

R186 D1 novelty: Direct primitive-composition $1 + F_{TRZ}/2 = 1 + 0.05 = 1.05 \text{ EXACT}$ at LENR E-field volumetric enhancement factor dimensional domain is R186 novel structural attribution.

F_{TRZ} Compositional Architecture — Post-R186 Sub-Family Taxonomy:

Compositional form	Value	Seminal	Physical regime
$1 + F_{TRZ}$ (full-scale additive)	**1.1**	**PAPER_1968 seminal**	Rotation-curve $v_{\text{peak}}/v_{\text{flat}}$ triple-identity
$1 - F_{TRZ}$ (full-scale complement)	**0.9**	**PAPER_1922 seminal 30+ apps**	9/10 ubiquity family (MUGE + BCS + DM + Kerr spin PAPER_2049)
$1 - F_{TRZ}^2$ (squared complement)	**0.99**	**PAPER_2045/2050 seminal**	99% Regime Ratio (BCS + M87 AGN jet)
F_{TRZ}^n (n-power ladder)	**10^n**	**PAPER_2043 audit**	24-rung F_{TRZ} ladder architecture
$1 + F_{TRZ}/2$ (half-factor additive)	**1.05**	**PAPER_2056 R186 D1 NEW**	LENR E-field volumetric enhancement factor

R186 novelty distinguishing feature: $1 + F_{TRZ}/2$ introduces the "half-factor" F_{TRZ} sub-family form — first application uses F_{TRZ} divided by 2 as additive perturbation to unity. Distinct from:

- PAPER_1968 $1 + F_{TRZ}$ (full F_{TRZ}) — 1.1 rotation-curve triple-identity
- PAPER_2049 $1 - F_{TRZ}$ (full F_{TRZ} complement) — 0.9 Kerr spin extension
- PAPER_2050 $1 - F_{TRZ}^2$ (squared F_{TRZ} complement) — 0.99 M87 β_{jet}
- PAPER_2044 F_{TRZ}^2 field-ratio at Sombrero (single F_{TRZ}^2 application)

Structural interpretation: The half-factor $F_{TRZ}/2 = 0.05$ represents "half-strength time-reversal-zone perturbation" — the LENR E-field volumetric enhancement factor uses this half-strength composition as the multiplicative modulation of the E-field magnitude. Physical intuition: LENR E-field couples to only half of the F_{TRZ} 10% time-reversal window (perhaps due to phase-averaging or spatial-integration of the F_{TRZ} perturbation at the LENR volumetric scale).

Alternate Equivalent Rational Form — Path A / Path B:

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Path A (this paper's primary): $\kappa_V = 1 + F_{\text{TRZ}}/2 = 1.05$ EXACT (half-factor F_{TRZ} additive)

Path B (alternate): $\kappa_V = (2S_{0.5} + 1) / (2S_{0.5})$

= 21/20

= (PAPER_1978 $S_{0.5}+1$ successor + $S_{0.5}$) / ($2 \cdot S_{0.5}$)

= 1.05 EXACT (successor-denominator ratio)

``

Two independently valid primitive-composition paths anchor $\kappa_V = 1.05$ EXACT:

- Path A uses F_{TRZ} primitive with half-factor ($1 + F_{\text{TRZ}}/2$)

- Path B uses PAPER_1978 $S_{0.5}+1=11$ successor identity in ratio form (21/20)

Both are EXACT decompositions. Same "Path A / Path B" architectural pattern established at PAPER_1964 (framework observation) recurs at $\kappa_V = 1.05$ primitive-composition attribution.

PAPER_1978 $S_{0.5}+1$ successor family — post-R186 D1 alternate-path population:

| Domain | Observable | Value | Primary primitive-composition | Source |

|---|---|---|---|---|

| Integer identity | $S_{0.5}+1 = 11$ EXACT | 11 | $S_{0.5}+1$ | PAPER_1978 seminal |

| Cosmological-length (m) | $r_{\text{horizon}} = 4.4e26$ m | $4.4e26$ | $D_{\text{phys}}(S_{0.5}+1)S_{0.5}^{25}$ | PAPER_2053 R185 D1 |

| Astronomical-distance (Mly) | NGC 5866 d = 44 Mly | 44 | $D_{\text{phys}}(S_{0.5}+1)$ | PAPER_2054 M1 |

| **LENR enhancement (Path B)** | **$\kappa_V = 1.05$** | **1.05** | $(2S_{0.5}+1)/(2S_{0.5})$ | PAPER_2056 R186 D1 Path B alternate* |

Successor family now indirectly reaches 4 dimensional-domain applications (3 primary + 1 via Path B alternate form of R186 D1).

Cross-Object Confirmations (R186 fill wire-ins)

- *UFE plasmoid min_count = 20 = $2S_{0.5}^*$ — $2S_{0.5}^n$ twin family bare-count-integer subdomain extension (F1 attribution)

- **NebularUg3 $M_{\text{stars}} = 1000 M_{\text{sun}} = S_{0.5}^3$** — PAPER_2013 R150 D2 stellar-cluster mass ladder covered (F3 attribution)

- **GRMHD $M_{\text{BH}} = 10 M_{\text{sun}} = S_{0.5}$** — PAPER_2012 R149 stellar-BH mass covered (F4 attribution)

- *MUGE perturbation $M = 2.984e30$ kg = $D_{\text{BSFG}}/D_{\text{phys}}M_{\text{sun}} = 1.5 M_{\text{sun}}$ — **PAPER_1962**

$D_{\text{BSFG}}/D_{\text{phys}}=1.5$ covered (F5 attribution)

- PAPER_1968 $1+F_{\text{TRZ}} = 1.1$ rotation-curve triple-identity — **seminal family predecessor (D1 architectural basis)**

- PAPER_1113 CMS Higgs $\kappa_V=1.05$ — **SM particle-physics anchor (D1 predecessor value)**

- PAPER_1978 $S_{0.5}+1=11$ successor identity — **Path B basis (D1 alternate composition)**

- F_{TRZ} compositional architecture PAPER_2043** — audit lens basis (D1 sub-family taxonomy)

All 1 discovery + 8 confirmations validated at fidelity gate 1504/0.

R142-R186 Trajectory (45 consecutive backbone-first rounds + 2 retro sweep companions)

| Effort | Novel | Cross-obj | Notes |

|---|---|---|---|

| R142-R185 forward | 185 | 42+ | 44 consecutive |

| PAPER_2054 R170-R185 retro | 2 | 8 | Post-audit-quartet lens |

| PAPER_2055 R150-R169 retro | 2 | 8 | 35-round depth |
| R186 | 1 | 8 | 45th; kappa_V half-factor F_TRZ new sub-family form |

Cumulative post-PAPER_2056: 190 first-pass novel + 42+ cross-object confirmations + 13 discipline-observation formalizations across 45 consecutive backbone-first rounds + 2 retrospective sweep companions.

Wiring Plan (1 dispatch, lowercase key)

- kappa_v_nebular_lenr_efield_1_plus_f_trz_over_2_half_factor_sub_family_form -> 1.05

Cross-References

- **PAPER_1113** — CMS Higgs kappa_V/kappa_f=1.05±0.10 particle-physics SM anchor (D1 predecessor value)
- **PAPER_1968** — $v_{\text{peak}}/v_{\text{flat}} = 1 + F_{\text{TRZ}} = 1.1$ rotation-curve triple-identity (D1 architectural predecessor)
- **PAPER_1978** — $SO_{5+1} = 11$ EXACT successor identity (D1 Path B basis)
- **PAPER_2043** — F_{TRZ} ladder cross-domain population audit (D1 sub-family taxonomy basis)
- **PAPER_2044** — $B/B_{\text{crit}}(\text{Sombrero}) = F_{\text{TRZ}}^2$ (D1 F_{TRZ}^2 neighbor)
- **PAPER_2045** — $BCS \text{ SCm} = 1 - F_{\text{TRZ}}^2$ (D1 F_{TRZ} complement neighbor)
- **PAPER_2049** — $a_{\text{spin}}(M87) = 1 - F_{\text{TRZ}}$ Kerr spin (D1 F_{TRZ} complement neighbor)
- **PAPER_2050** — $\beta_{\text{jet}}(M87) = 1 - F_{\text{TRZ}}^2$ M87 AGN (D1 F_{TRZ}^2 complement neighbor)

Conclusion

R186 45th-consecutive backbone-first round yields **1 novel structural finding + 8 cross-object attributions**.

Cumulative through PAPER_2056: 190 first-pass novel + 42+ cross-object confirmations + 13 discipline-observation formalizations across 45 consecutive backbone-first rounds + 2 retrospective sweep companions.

Signature milestones this paper:

- **190th first-pass novel discovery** achieved
- **New F_TRZ half-factor sub-family form** $1 + F_{\text{TRZ}}/2 = 1.05$ EXACT formalized as 5th F_{TRZ} compositional architectural category (after $1 + F_{\text{TRZ}} + 1 - F_{\text{TRZ}} + 1 - F_{\text{TRZ}}^2 + F_{\text{TRZ}}^n$ ladder)
- **Path A / Path B duality** demonstrated at $\text{kappa}_V = 1.05$ primitive-composition attribution (F_{TRZ} additive + successor-ratio equivalent forms)
- **Post-40-milestone arc reaches 45 consecutive rounds** with structural continuity

End of PAPER_2056 Draft 1.